



MAHESH SELLAMUTHU

PRINCIPAL / SOLUTION ARCHITECT | CLOUD, AI & ENTERPRISE SOLUTIONS

Azure | .NET | Microservices | Cloud Architecture | Generative AI | Agentic AI |

RAG | MCP | DevOps | Distributed Systems

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PROFESSIONAL SUMMARY

Principal Architect with 15+ years of experience designing scalable cloud-native enterprise applications and distributed platforms using C#, .NET and Microsoft Azure, including 8+ years in architecture and technical leadership roles.

Architecture & Platforms: Enterprise architecture, cloud-native microservices, distributed systems, and high-availability design for platforms serving 10M+ users.

AI & Emerging Tech: Generative AI and Agentic AI solutions built on LLMs, Retrieval-Augmented Generation (RAG), vector databases, and MCP-based tool integration using Azure AI Foundry and Azure OpenAI.

Engineering Impact: 99.99% uptime, up to 40% performance improvement, 20% infrastructure cost savings, and 50% faster delivery through DevOps automation and CI/CD excellence.

Leadership & Delivery: Technical leadership, Agile delivery, security governance, and stakeholder management; mentoring engineers and architects and owning multi-year technology roadmaps.

Target Roles: Lead Technical Architect, Principal Architect, Technical Architect, Solution Architect, and Cloud/AI Architect roles in cloud-native and AI-driven product and enterprise organizations.

CORE SKILLS

AI & LLM Engineering: Generative AI, Agentic AI, Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Vector Databases (Azure AI Search, pgvector, Pinecone), Model Context Protocol (MCP), Microsoft AI Foundry / Azure AI Foundry, Azure OpenAI, Embeddings & Semantic Search, Prompt Engineering, AI Agents & Orchestration, LLM Cost & Performance Optimization

Cloud & Platform: Microsoft Azure, Azure Architecture, Azure IaaS, Azure PaaS, Azure Functions, Azure Service Bus, Azure Search, Azure Key Vault, Azure IoT Hub

Architecture & Design: Enterprise Architecture, Microservices Architecture, Cloud-Native Design, Distributed Systems, System Design, High Availability, Scalability, Disaster Recovery

Backend Development: C#, .NET Core, ASP.NET, Python, Entity Framework, Web API, REST APIs

DevOps & Automation: Azure DevOps, CI/CD Pipelines, DevOps Automation, Docker, Kubernetes, Containerization

Databases & Data Stores: SQL Server 2022, PostgreSQL, Redis, Elasticsearch

Security & Governance: Cloud Security, AI Security, Security Architecture, AI Governance, AI Guardrails, Identity & Access Management, OAuth 2.0, JWT, OpenID Connect, Microsoft Entra ID / Azure AD, Key Vault, API security, Zero Trust, RBAC, Managed Identity, Workload Identity, Compliance & Governance

Leadership & Delivery: Technical Leadership, Agile, Scrum, Stakeholder Management, Performance Optimization, API Strategy, System Integration

PROFESSIONAL EXPERIENCE

Principal Architect | Wissen Technology, Bangalore | Dec 2024 – Present

- Defined enterprise architecture roadmap for a Battery Swapping & Energy Platform, enabling multi-region deployments, 99.99% availability, and multi-currency billing.
- Architected cloud-native microservices platforms on Azure to support high throughput, fault tolerance, and future scale.
- Delivered 30% performance improvement and 20% infrastructure cost savings (~₹80L/year) through architecture optimization and cloud cost governance.
- Established CI/CD standards and DevOps automation, accelerating delivery speed by 50% across teams.
- Acted as senior technology advisor to business stakeholders, ensuring alignment between technology strategy and business outcomes.
- Designed and introduced enterprise GenAI/RAG architecture using vector databases and semantic search over operational and knowledge data, with MCP-based tool integration to securely connect AI assistants with enterprise systems.

Technical Architect | RGSI, Bangalore | Jul 2024 – Oct 2024

- Led architecture and delivery of the EQLMS enterprise platform, streamlining system integrations.
- Improved backend processing and API performance by 25% through architectural and data-flow optimization.
- Ensured scalability, security, and maintainability using cloud-native best practices.

Assistant Director – Technology | Ernst & Young (EY), Bangalore | Nov 2021 – Apr 2024

- Led enterprise-wide architecture modernization initiatives for platforms serving 10M+ users.
- Improved application performance by 40% and usability metrics by 25%.

- Drove Agile and DevOps transformation for a 20-member engineering organization, reducing release cycles from 6 weeks to 2 weeks.
- Partnered with risk and security teams to ensure enterprise-grade compliance, governance, and cloud security.
- Mentored senior engineers and architects, strengthening technical standards and leadership capability.

Senior Software Engineer | Allscripts, Bangalore | Sep 2018 – Oct 2021

- Developed scalable healthcare systems processing 100K+ daily patient records.
- Increased platform reliability to 99.9% uptime through performance tuning and resilient design.
- Reduced application latency by 40% using optimized APIs, database tuning, and caching strategies.

Associate Team Lead | Zenith Software Pvt. Ltd., Bangalore | Jan 2015 – Sep 2018

- Delivered a global hotel booking platform integrated with Expedia and third-party APIs.
- Improved booking speed by 30% and enabled 50% transaction growth.
- Led engineering efforts and ensured consistent delivery quality.

Developer | Interface Business Solutions Pvt. Ltd., Mumbai | Jun 2013 – Nov 2014

- Built and integrated e-commerce systems, increasing transaction throughput by 3x.
- Reduced defects by 40% through structured code reviews and best practices.

Senior Software Engineer | Zobble Solutions Pvt. Ltd., Mumbai | Nov 2011 – May 2013

- Developed e-learning platforms supporting 20K+ users.
- Achieved 35% faster feature delivery through reusable components and improved workflows.
- Managed a 10-member development team under tight deadlines.

Software Consultant | Info Jyothi Software Consultancy, Chennai | Oct 2010 – Oct 2011

- Delivered applications for PWD government projects, reducing manual workflows by ~50% and improving operational efficiency.

EDUCATION

Bachelor of Engineering – Computer Software Engineering | Nandha Engineering College, Anna University | 2010 | 70.18%

LANGUAGES

English | Tamil | Hindi